

DANIEL R. FRITCH

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Key Skills:

- Thermal-fluid system design, 3D solid modeling, finite element analysis

EDUCATION

Villanova University: College of Engineering – Villanova, Pennsylvania

Bachelor of Science in Mechanical Engineering, 05/2013 (GPA: 3.67)

- Concentration in thermal-fluid systems

Master of Science in Mechanical Engineering, Expected 05/2015 (Current GPA: 3.42)

King's College London – London, UK, 01/2012 to 06/2012

HONORS/AWARDS

Dean's Award for Academic Excellence, Pi Tau Sigma Mechanical Engineering Honor Society, Tau Beta Pi Engineering Honor Society, W.W. Smith Scholarship, Villanova Undergraduate Research Fellowship, National Society of Collegiate Scholars, Villanova Academic Scholarship

WORK/RESEARCH EXPERIENCE

Graduate Research Assistant, *Villanova University: Laboratory for Advanced Thermal and Fluid Systems, Villanova, PA*, 09/2013 to Present

- Designed, built, and tested a two-phase flow loop for microchannel evaporator characterization
- Created a conjugate heat transfer model for optimizing evaporator geometry
- Hands-on machining experience

Assistant Engineer, *Materials Research and Design, Wayne, PA*, 06/2012 to 08/2013

- Used finite element analysis to assist in design of thermal protection system components for hypersonic vehicles

Undergraduate Research Fellow, *Villanova University: Biothermal Sciences Laboratory, Villanova, PA*, 05/2011 to 12/2011

- Investigated the mechanics of homogeneous nucleation in micro-scale water droplets as it applies to cryogenic freezing of cells
- Developed testing procedures and optimized materials for creating water-in-oil micro-emulsions.

TECHNICAL KNOWLEDGE

Software: SolidWorks, AutoCAD, MATLAB, MathCAD, ANSYS Traditional, ANSYS Workbench, ANSYS FLUENT, COMSOL, Abaqus CAE, and all Microsoft Office products

References for Daniel Fritch

Alfonso Ortega, Ph.D.

Associate Vice-President for Research and Graduate Programs

Director of the Laboratory for Advanced Thermal and Fluid Systems

Villanova University

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Brian J. Sullivan, Ph.D.

Director

Materials Research and Design

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